COMPLEX POWERS, REAL ANSWERS

From i^{i} to ω^{ω}

One rule decides when a power of a unit-circle number lands back on the real line.

THE RULE

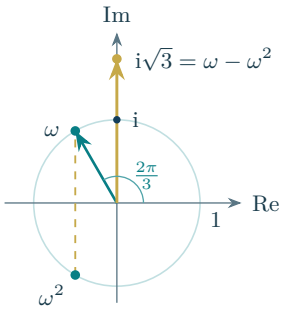
Imaginary part sets the size, real part sets the angle

Take a base on the unit circle, $u = \mathrm{e}^{\mathrm{i}\theta}$, with a chosen argument θ, so that $\operatorname{Log} u = \mathrm{i}\theta$ and $u^w := \mathrm{e}^{w \operatorname{Log} u}$. Writing the exponent as $w = a + \mathrm{i}b$,

$$u^{a+\mathrm{i}b} \;=\; \underbrace{\mathrm{e}^{-b\theta}}_{\text{size}} \; \underbrace{\mathrm{e}^{\mathrm{i}a\theta}}_{\text{angle}}$$

so the result is real exactly when $a\theta \in \pi\mathbb{Z}$, positive when $a\theta/\pi$ is even and negative when it is odd.

base	θ	real when
i	$\pi/2$	$a \in 2\mathbb{Z}$
ω	$2\pi/3$	$a \in \frac{3}{2}\mathbb{Z}$, sign $(-1)^{2a/3}$



CASE $a = 0$: PURELY IMAGINARY EXPONENTS GIVE POSITIVE REALS

i^{i}, ω^{i} and $\omega^{1+2\omega}$

With no real part in the exponent there is no rotation at all, so every one of these is a positive real; the third needs only the observation that doubling ω and adding 1 cancels its real part, $1 + 2\omega = \omega - \omega^2 = \mathrm{i}\sqrt{3}$.

$$\mathrm{i}^{\mathrm{i}} = \mathrm{e}^{\mathrm{i}\cdot\mathrm{i}\pi/2} = \mathrm{e}^{-\pi/2} \approx 0.207880, \qquad \omega^{\mathrm{i}} = \mathrm{e}^{-2\pi/3} \approx 0.123145, \qquad \omega^{1+2\omega} = \mathrm{e}^{\mathrm{i}\sqrt{3}\cdot\mathrm{i}(2\pi/3)} = \mathrm{e}^{-2\pi/\sqrt{3}} \approx 0.0265799.$$

CASE $a \neq 0$

Powers of ω^{ω}

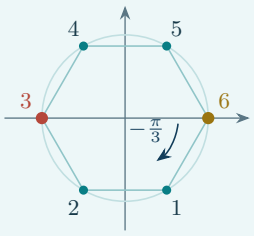
Since $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$, the exponent ω has $a = -\frac{1}{2}$, so

$$\omega^{\omega} = \mathrm{e}^{\omega\cdot\mathrm{i}(2\pi/3)} = \mathrm{e}^{-\pi/\sqrt{3}}\,\mathrm{e}^{-\mathrm{i}\pi/3}.$$

For an integer n the power $(\omega^{\omega})^n = \omega^{n\omega}$ has $a = -n/2$, so its angle is $-n\pi/3$ and it is real exactly when $3 \mid n$, negative for odd multiples of 3 and positive for even ones.

$$\boxed{\omega^{3\omega} = -\mathrm{e}^{-\pi\sqrt{3}}} \approx -0.00433342$$

$$\boxed{\omega^{6\omega} = \mathrm{e}^{-2\pi\sqrt{3}}} \approx 1.87785 \times 10^{-5}$$



Direction of $\omega^{n\omega}$; each step also shrinks the size by $\mathrm{e}^{-\pi/\sqrt{3}} \approx 0.163$.

Where the three comes from

The step angle is $a\theta = \operatorname{Re}(\omega) \cdot \frac{2\pi}{3} = -\frac{\pi}{3}$, so the three is inherited from ω being a cube root of unity. The same recipe for i gives $\operatorname{Re}(\mathrm{i}) \cdot \frac{\pi}{2} = 0$, which is why i^{i} needs no power at all, and for $\zeta = \mathrm{e}^{2\pi\mathrm{i}/m}$ the step is $\frac{2\pi}{m}\cos\frac{2\pi}{m}$.

An irrational component

The real number $\sqrt{3}$ is irrational. The number $\mathrm{i}\sqrt{3} \notin \mathbb{Q}$ is **purely imaginary and algebraic**, with irrational imaginary part: $(\mathrm{i}\sqrt{3})^2 + 3 = 0$. In standard terminology, "irrational" describes real numbers. Notice that $\mathrm{i}\sqrt{3} = 1 + 2\omega$ is nevertheless an **Eisenstein integer**.

ANY BASE, WITH THE NATURAL LOG

Read the answer through $\ln$

For a base $z = r\mathrm{e}^{\mathrm{i}\theta}$ the logarithm is $\ln z = \ln r + \mathrm{i}\theta$, and since $z^w = \mathrm{e}^{w\ln z}$ with $w = a + \mathrm{i}b$, the product $w \ln z = (a\ln r - b\theta) + \mathrm{i}\,(b\ln r + a\theta)$ has real part $\ln|z^w|$ and imaginary part equal to the angle, so z^w is real exactly when $\operatorname{Im}(w\ln z) \in \pi\mathbb{Z}$. On the unit circle $\ln r = 0$ and this is the rule above; off it, the imaginary part of the exponent also turns the result through $b\ln r$, so $2^{\mathrm{i}} = \mathrm{e}^{\mathrm{i}\ln 2}$ is not real while $2^{\mathrm{i}\pi/\ln 2} = \mathrm{e}^{\mathrm{i}\pi} = -1$.

Logarithm convention. Values use the principal logarithms $\operatorname{Log}\mathrm{i} = \mathrm{i}\pi/2$ and $\operatorname{Log}\omega = \mathrm{i}(2\pi/3)$. On another branch, $\theta \mapsto \theta + 2\pi k$, the numbers change: i^{i}, ω^{i} and $\omega^{1+2\omega}$ stay positive, $\omega^{3\omega}$ stays real with sign $(-1)^{k+1}$, and integer powers such as $(\omega^{\omega})^n = \omega^{n\omega}$ remain valid on every branch.